

TriVerse AI: Handwritten Character Recognition Report

Author: CodeAlpha Candidate / TriVerse Platform Core

Date: June 2026

Executive Summary

This report documents the computer vision pipeline within TriVerse AI's VisionVerse, focusing on handwritten character recognition. Using PyTorch, we implemented and compared a Custom CNN and a ResNet18 model. Models were tested on MNIST digits and the EMNIST balanced set containing characters and letters.

Dataset Description

We used the standard MNIST dataset (70,000 28x28 grayscale images of digits 0-9) and the EMNIST Balanced dataset (131,600 images representing 47 balanced classes of digits and upper/lower case letters).

Data Cleaning & Augmentation

Images were normalized using channel mean and standard deviation. We applied random affine transformations, rotations (up to 10 degrees), and scaling in PyTorch to augment the training data, helping the models handle varied handwriting styles in the drawing canvas.

Model Training

A Custom CNN was built with three convolutional layers, max pooling, dropout (0.25), and a dense layer. ResNet18 was adapted for grayscale input and fine-tuned on GPU compute. Training ran for 10 epochs using an Adam optimizer and Cross-Entropy loss.

Results & Comparisons

On MNIST, ResNet18 achieved 99.4% Accuracy, while the Custom CNN reached 98.9%. On the complex EMNIST dataset, ResNet18 reached 91.2% accuracy, outperforming the Custom CNN's 87.8%. The Custom CNN is significantly faster (sub-2ms inference, 12MB size) compared to ResNet18 (6ms inference, 45MB size).

Canvas Inference and Explainability

The system includes an interactive canvas drawing interface. Predictions are made in real-time. Activation maps (Grad-CAM style) are computed to highlight pixel-density groups that contributed most to the model's classification output.

Conclusions & Future Work

Both models have been integrated successfully. For lightweight browser/desktop clients, the custom CNN is preferred due to low latency. Future work will focus on offline optical character recognition (OCR) parsing whole sentences rather than individual characters.